

Skills Summary

Reading and Sketching Slope Fields

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading a Slope from the Field

What a field is. At each lattice point (x, y) the picture draws a short segment whose slope is the value of $\frac{dy}{dx}$ there. Nothing is solved — the equation is simply evaluated, one point at a time.

Two tells. Parallel segments across a *row* mean the slope depends only on y ; parallel down a *column* means it depends only on x .

Example: For $\frac{dy}{dx} = x + y$, find the slope at $(2, 1)$.

Step 1. A slope field asks nothing more than the value of the right-hand side, so substitute the coordinates straight into the equation.

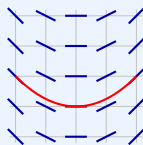
Step 2. $\frac{dy}{dx} = 2 + 1$
 $\Rightarrow 3$

Try it: Same equation. Find the slope at $(3, -1)$.

2 :check

Skill 2 — Sketching a Solution Curve

Rule. Start at the given point and move so the curve is always *tangent* to the segments beneath it. Never cut across a segment, and never let two solution curves touch.



The red curve is one solution of $\frac{dy}{dx} = \frac{x}{2}$: flat where the segments are flat, steeper where they are steeper.

Example: To sketch a solution of $\frac{dy}{dx} = \frac{x}{2}$, find the slopes at $x = 2$ and $x = 4$.

Step 1. A sketch is only as good as the slopes it is built from, so read a few off the equation before drawing anything.

Step 2. $\frac{2}{2}$ and $\frac{4}{2}$
 $\Rightarrow 1$ and 2 : the curve doubles its steepness between them.

Try it: For $\frac{dy}{dx} = \frac{x}{3}$, find the slopes at $x = 3$ and $x = 6$.

check: 1, 2

Skill 3 — From the Field to a Formula

When the slope depends only on x , the equation is just an antiderivative: $\frac{dy}{dx} = f(x)$ gives $y = F(x) + C$, and every solution curve in the field is a vertical shift of one another. A point on the curve fixes C : substitute the coordinates and solve.

Example: Solve $\frac{dy}{dx} = 3x^2$ through the point $(1, 5)$.

Step 1. The slope depends on x alone, so antidifferentiate first and let the given point pin down the constant.

Step 2. $y = x^3 + C$, and $5 = 1 + C$.

$\Rightarrow \mathbf{y = x^3 + C}$ with $\mathbf{C = 4}$.

Try it: Solve $\frac{dy}{dx} = 4x^3$ through $(1, 7)$.

check: 1, 2